



# Articulate AI

AI-Powered Spoken English Coach  
*Built for Bangladesh. Built for Fluency.*

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CUET AI Hackathon — Project Report · 2026

**Team** articulate-ai

Pritom Biswas · Moumita Sarkar Munny · Anirban Biswas · Syeda Afrin Jahan

## THE PROBLEM

# Bangladeshi English Learners Are Stuck

1

### No Speaking Feedback

Local IELTS courses teach grammar — speaking is never evaluated. Students graduate without ever being scored on pronunciation.

2

### Global Apps Don't Fit Bengali Learners

Fixed pace, Western-centric content. Bangla phonology lacks 7 English phonemes (/v/, /æ/, /θ/, /ð/, /z/, /ɪ/, /w/) — existing apps never address this.

3

### Prohibitively Expensive

Private tutor: BDT 8,000–24,000/month · Coaching centre: BDT 4,000–8,000/month · IELTS exam fee alone: BDT 28,450

4

### Zero Personalisation

Same content for every learner. No adaptation to CEFR level, pace, weak phonemes, or native-language context. One-size-fits-all fails everyone.

## OUR SOLUTION

# Articulate AI — The Complete Spoken English Ecosystem

*The only AI-powered spoken English coach built for Bengali speakers — from placement test to IELTS band score*

### Full Learning Ecosystem

Onboarding placement → vocabulary → pronunciation → IELTS → exams — all in one platform

### Unbeatable Price

BDT 200–500/month vs. BDT 8K–24K private tutor · AI infra ~BDT 2.5/call · 70–85% gross margin at scale

### Personal AI Guidance

YOUR weak words, YOUR CEFR level, YOUR pace — SM-2 spaced repetition tied to pronunciation accuracy

### Phoneme-Level AI Scoring

Azure IPA granularity + 7 Bangla-specific risk phonemes + native Bangla feedback tips every session

### IELTS Simulation

Voice AI examiner · scored turns · IELTS band 4–9 report with per-turn breakdown and Bangla summary

### Gamified & Sticky

XP · streaks · level system · achievement badges · leaderboard — SRS flashcards drive daily return

## 8 Integrated Modules



### Onboarding

Mic check · CEFR test · 3 attempts · A1–C1 placement



### Curriculum

8 chapters · 40 lessons · Oxford 3000 · CEFR A1–C1



### Pronunciation

Azure IPA · phoneme chips · Bangla tips · score history



### IELTS Talk

Voice AI examiner · scored turns · band report



### Exam System

AI-generated · Listening + Speaking · async eval



### Vocab & SRS

SM-2 flashcards · bookmarks · audio cache



### Progress & XP

Streaks · badges · leaderboard · 7-day chart



### AI Chat Tutor

GPT-4o-mini · grammar · word lookup · TTS

COST & MARKET

# Learners Save More. We Earn More.

How We Compare

| Platform        | Monthly Cost | Speaking Feedback  | Bangla Support |
|-----------------|--------------|--------------------|----------------|
| Private Tutor   | BDT 8K–24K   | Human · subjective | ✓ Yes          |
| Coaching Centre | BDT 4K–8K    | Minimal · group    | Part.          |
| Online Course   | BDT 150–800  | None               | ✓ Yes          |
| ELSA Speak      | BDT ~1,440   | AI · word-level    | ✗ No           |
| Duolingo Max    | BDT ~3,600   | Minimal            | ✗ No           |
| Articulate AI   | BDT 200–500  | AI · phoneme-level | ✓ Yes (বাংলা)  |

HUMAN-LED

Instructor cost: BDT 30K–60K/month  
Scaling: linear with headcount  
Gross margin: 20–30%  
Availability: 8–12 hrs/day

AI-LED ✓ ARTICULATE AI

Infra cost: ~BDT 2.5 per API call  
Scaling: near-zero marginal cost  
Gross margin: 70–85%  
Availability: 24/7

Sources: British Council BD · ieltsbd.co · pricetodayhub.com · ELSA App Store · Duolingo public pricing · Author survey 2025

Market Opportunity

BDT 490 Cr+

Addressable Market

English-learning edtech in Bangladesh · Freemium SaaS + institutional B2B

160M+

Bengali Speakers

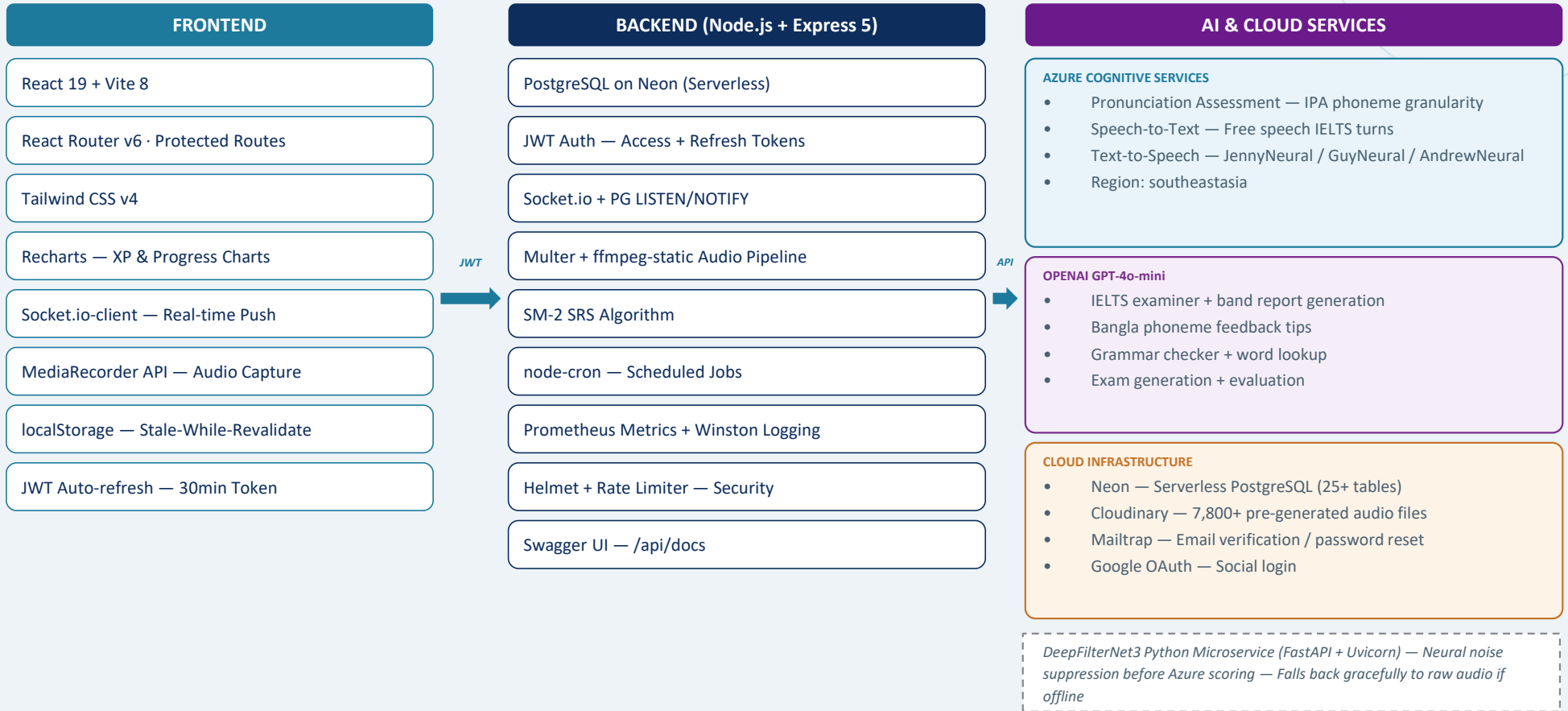
Primary: 16–35 age group in Bangladesh · Secondary: diaspora, West Bengal

BDT 200–500

Target Monthly Price

96% cheaper than private tutor · 65% cheaper than ELSA Speak · Full Bangla support

# Technical Architecture



## WHAT MAKES US DIFFERENT

# AI Innovation Deep-Dive

## PRONUNCIATION PIPELINE

1. User records via browser MediaRecorder (webm/mp4/ogg)
2. Audio routed to DeepFilterNet3 Python microservice — neural noise removal
3. ffmpeg converts to 16kHz mono WAV → Azure Pronunciation Assessment (IPA granularity)
4. Returns: overall\_score, accuracy, fluency, completeness + per-phoneme IPA accuracy
5. 7 Bangla-risk phonemes flagged (/v/, /æ/, /θ/, /ð/, /z/, /ɪ/, /w/)
6. Bangla tip generated by GPT-4o-mini in background (fire-and-forget)

Azure Cognitive Services · DeepFilterNet3 · GPT-4o-mini

## IELTS CONVERSATION AI

Turn-by-turn voice conversation with AI examiner (GPT-4o-mini)  
Two-pass free speech assessment: Azure STT gets transcript → re-runs pronunciation assessment with that transcript as reference — no pre-known text required  
Band report: structured JSON → IELTS band 4–9 (0.5 steps), per-turn breakdown, mispronounced words, fluency issues, Bangla one-line summary  
75 XP awarded on session completion

GPT-4o-mini · Azure STT + Pronunciation · Azure Neural TTS

## SM-2 SPACED REPETITION

Standard SM-2 maps a 0–5 quality rating to review intervals.=  
Our innovation: pronunciation accuracy score (0–100%) IS the quality parameter — not subjective recall confidence.  
Score ≥80% = quality 5 (long interval) · Score <40% = quality 1 (back next day)  
Word states: NEW → LEARNING (streak ≥1) → FAMILIAR (streak ≥3) → MASTERED (streak ≥6)  
Due cards served via PostgreSQL view vw\_due\_cards, capped at 20/session

## BANGLA-NATIVE AI FEEDBACK

Every score, tip, badge, exam report, error message delivered in Bengali (বাংলা)  
banglaRiskPhonemes.js — 7 IPA phonemes Bengali speakers systematically mispronounce, each with Bangla articulation instructions  
OpenAI generates one cohesive Bangla tip targeting the user's specific weak phonemes from that session  
IELTS band report includes a one-sentence Bangla summary for instant comprehension

## LLM-POWERED CONTENT

Oxford 3000 curriculum rebuilt from PDF using a custom column-aware parser (4-column layout)  
7,640 words · Bangla meanings · IPA · CEFR levels · frequency ranks  
160 lesson phrases generated by GPT-4o-mini using each lesson's exact vocabulary — contextual reinforcement  
All word and phrase audio pre-generated via Azure Neural TTS, stored on Cloudinary (7,800+ files)

# Eight Modules. One Platform. Unbeatable Value.

## Integrated Learning Modules

**Onboarding**  
Mic check · CEFR placement · 3 attempts · A1–C1

**Curriculum**  
8 chapters · 40 lessons · Oxford 3000 · CEFR A1–C1

**Pronunciation**  
Azure IPA · phoneme chips · Bangla tips · sparkline

**IELTS Talk**  
Voice AI examiner · scored turns · band 4–9 report

**Exam System**  
AI-generated tests · Listening + Speaking · async eval

**Vocab & SRS**  
SM-2 flashcards · bookmarks · 7-day TTL audio cache

**Progress & XP**  
Streaks · badges · leaderboard · 7-day XP chart

**AI Chat Tutor**  
GPT-4o-mini · grammar check · word lookup · profile-aware

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LOOKING FORWARD

# Impact, Roadmap & Team

## 7,640

Oxford 3000 Words

with IPA, Bangla meaning & Azure TTS audio

## 8

AI Services Integrated

Azure STT · TTS · Pronunciation · OpenAI GPT-4o-mini ·  
DeepFilterNet3 · Neon · Cloudinary · Socket.io

## BDT 490Cr+

Addressable Market

English-learning edtech in Bangladesh · Freemium SaaS +  
institutional B2B

## Roadmap

NOW

Full platform live · 8 modules · Azure +  
GPT-4o-mini · CEFR A1–C1

Q3 2026

Mobile app (React Native) · Android-first  
BD market · Offline mode

Q1 2027

Institutional B2B · School/university API ·  
Admin dashboards

2027+

West Bengal expansion · Hindi/Urdu  
phoneme modules · Multi-dialect AI

## The Team — CUET AI Hackathon 2026



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